

WORKSHEET A.5.M4

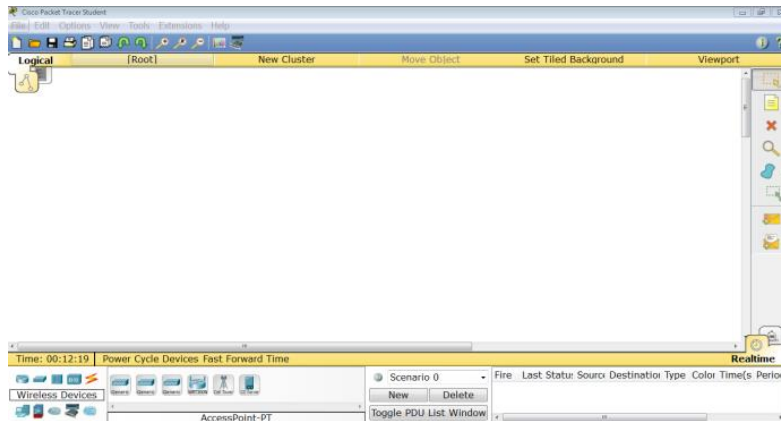
Wireless Networks

What we will learn today:

- Build wireless computer networks.
- Apply basic wireless-network security settings.
- Identify wireless-network problems and propose and apply solutions.

Consolidation Activities

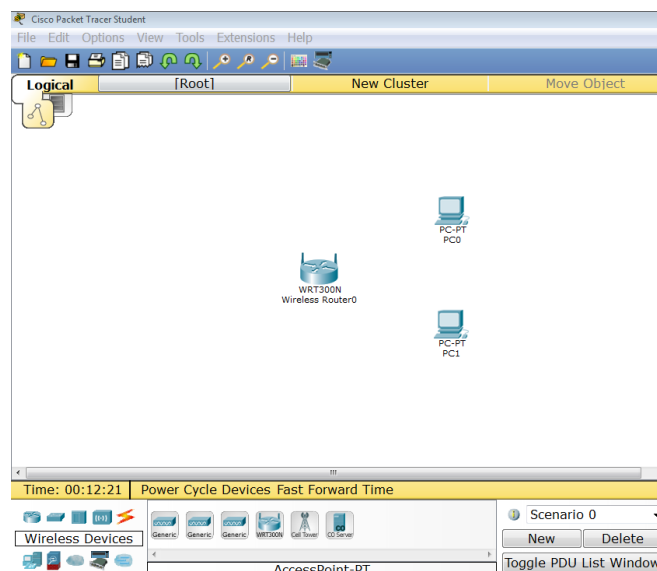
1. Open Cisco Packet Tracer.



2. Design a wireless network using:

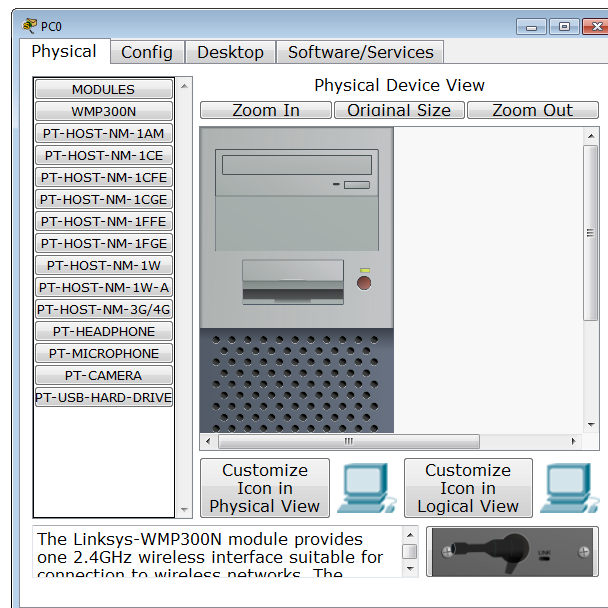
- (a) One wireless router.
- (b) End devices (PCs or laptops) equipped with wireless network cards.

3. Drag a wireless networking device to the centre of the workspace.

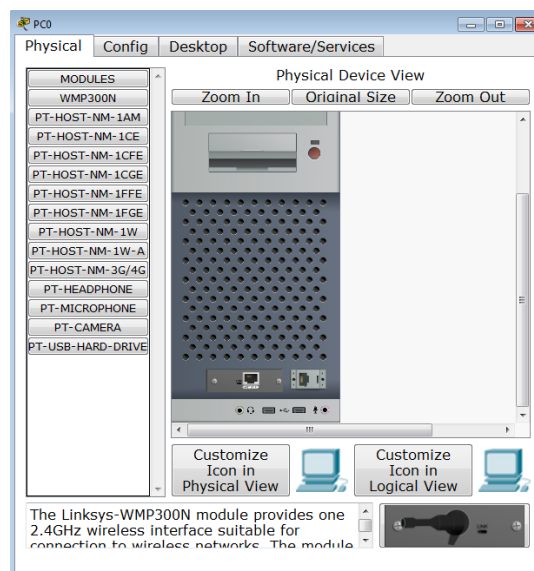


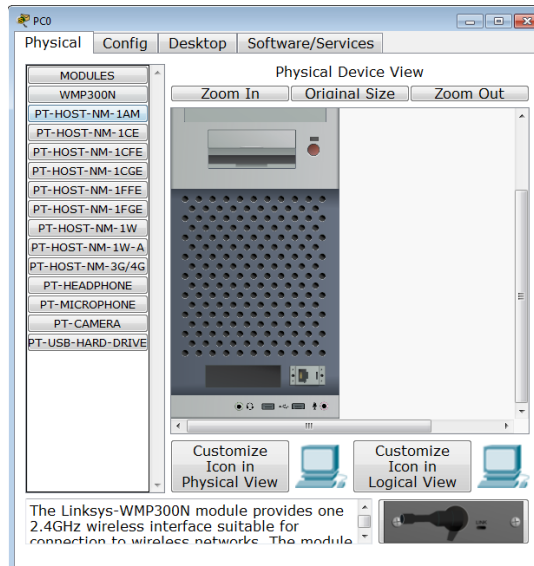
Choose Wireless Devices, then the WRT300N model.

4. Add two computers, PC0 and PC1.
5. Equip both computers with wireless network cards so that they can connect to the wireless router:
 - (a) Select PC0 and switch it off using the ON/OFF button.
 - (b) Confirm that the light above the switch changes from green to black.



- (c) While the computer is off, remove its network card by dragging it to the lower-right area of the window.
- (d) Confirm that an empty black slot appears.
- (e) Drag the wireless network card from the lower-right area into the empty slot.



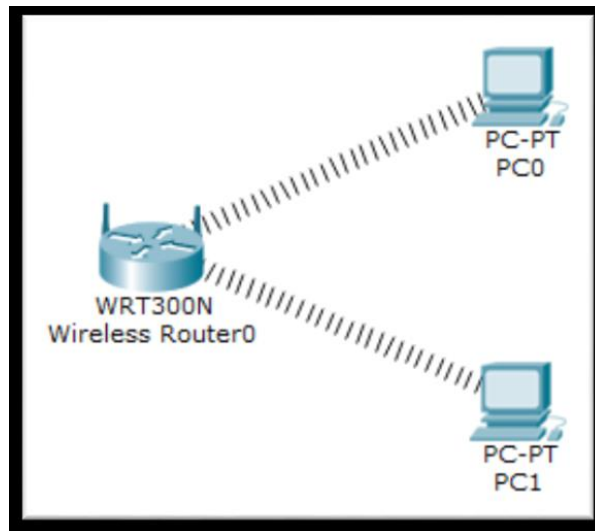


(f) Turn PC0 on using the ON/OFF button.

(g) Close the PC0 window.

(h) Open PC1 and repeat the procedure to install a wireless network card.

The completed wireless network should resemble:



6. Record the IP address assigned by the router to each PC.

PC0: ____ . ____ . ____ . ____

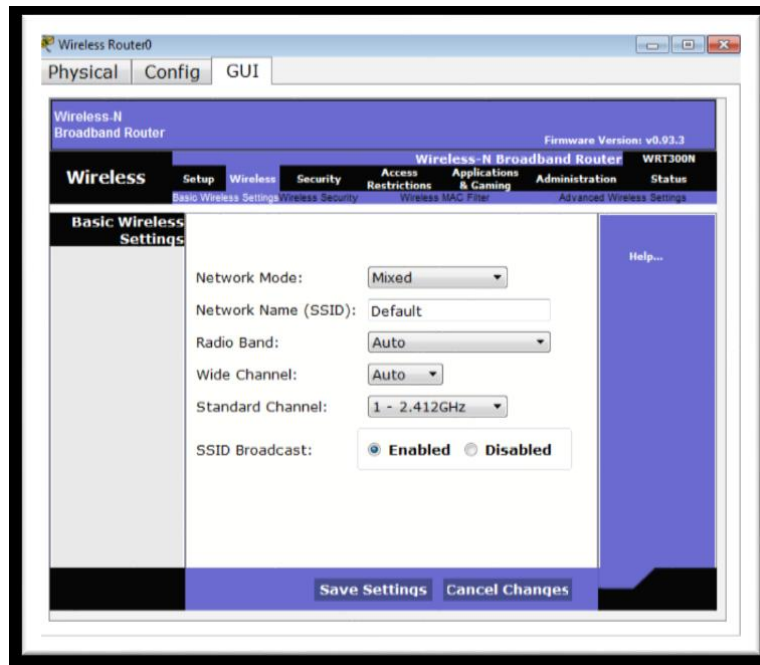
PC1: ____ . ____ . ____ . ____

7. Use ping to check whether PC0 communicates with PC1. Identify and solve any problems.

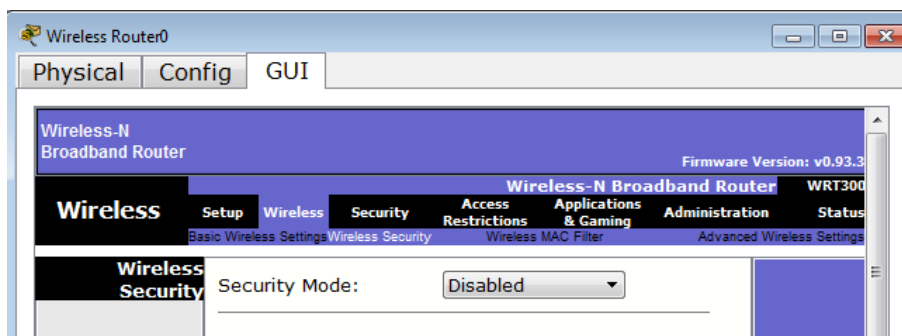
WIRELESS NETWORK SECURITY

8. The network you created has a security weakness: any newly added computer can connect without a password. Secure the wireless router as follows:

- (a) Select the WRT300N router and choose the GUI tab.
- (b) Select Wireless. Change the Network Name (SSID) from Default to SCHOOL, then select Save Settings.



- (c) Select Wireless Security and change Security Mode from Disabled to WPA2 Personal.



Disabled means an open network without a password. WEP, WPA Personal and WPA2 Personal provide password protection.

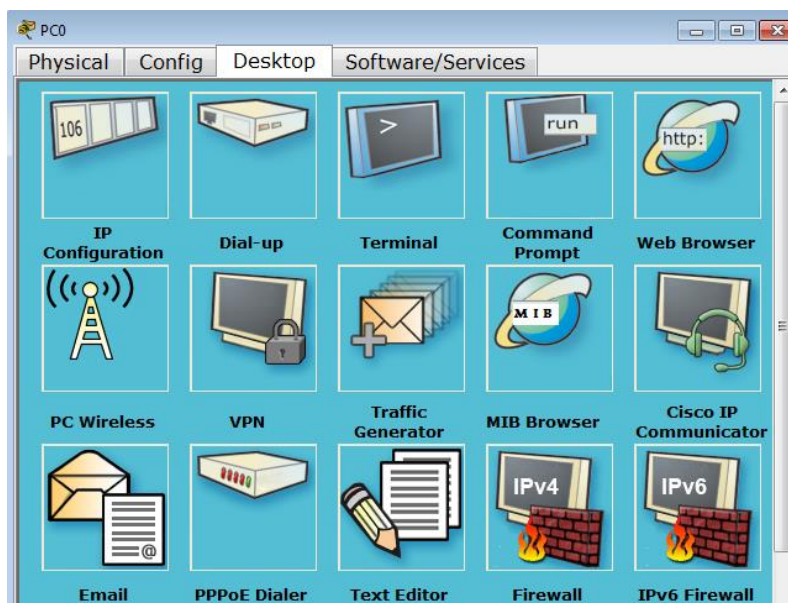
- (d) In the Passphrase field, enter a password of at least eight characters.
- (e) Enter guesswhat and select Save Settings.



After changing the network name (SSID) and password, the network is reasonably secure.

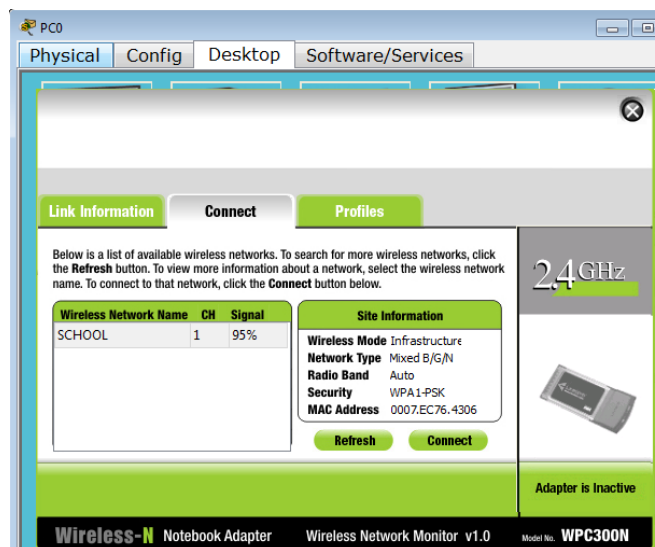
9. The two computers have lost their connection to the wireless router and must now be configured with the new security settings:

(a) Select PC0, choose the Desktop tab and select PC Wireless.



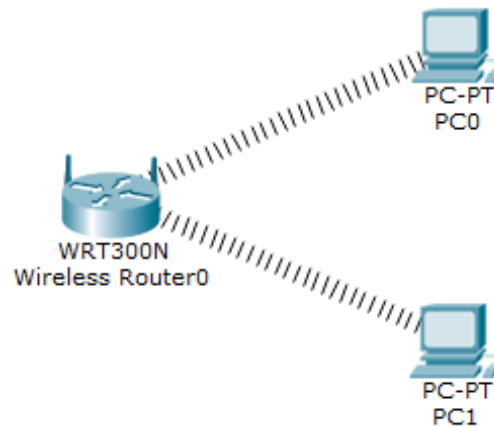
(b) Select the Connect tab and press Refresh until the SCHOOL network appears.

(c) Select SCHOOL and press Connect.



- (d) Enter guesswhat in the Pre-shared Key box and select Connect.
- (e) Close the two open windows and confirm that PC0 has reconnected to the router.
- (f) Repeat the process for PC1.

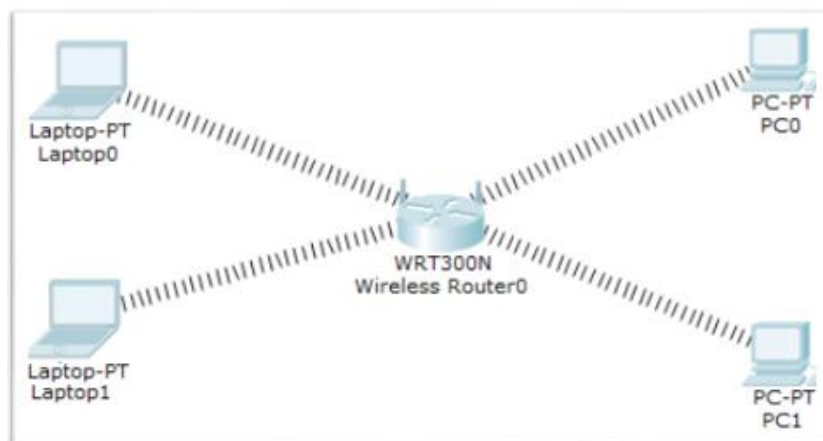
The final network should look like this:



10. Save the file using File > Save (or the disk icon) as Network2 in the folder specified by your teacher.

Additional Activities

11. Create the network shown below.



- 12. Open Troubleshooting 2.pka and find the errors in the wireless network.
- 13. Open Troubleshooting 3.pka and follow the displayed instructions.
- 14. With your teacher, implement this activity in a real environment using a small wireless router and laboratory computers fitted with USB wireless adapters.